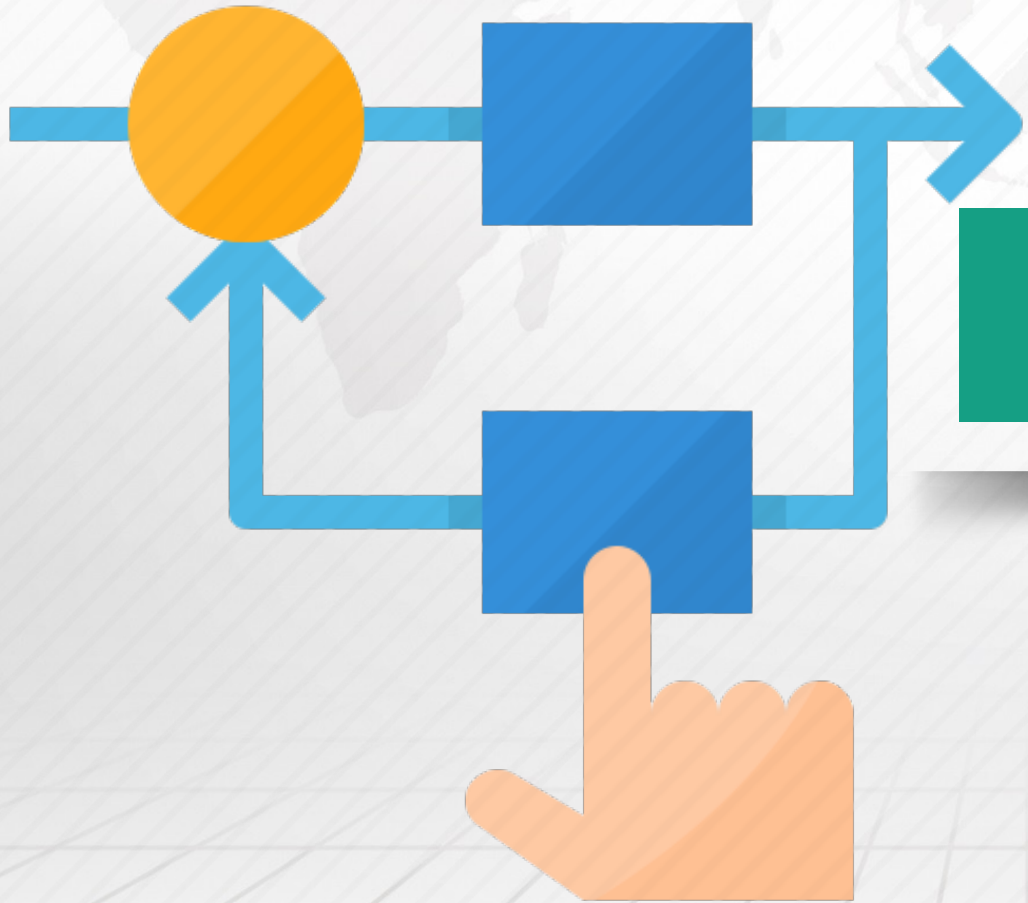




江苏科技大学
JIANGSU UNIVERSITY OF SCIENCE AND TECHNOLOGY



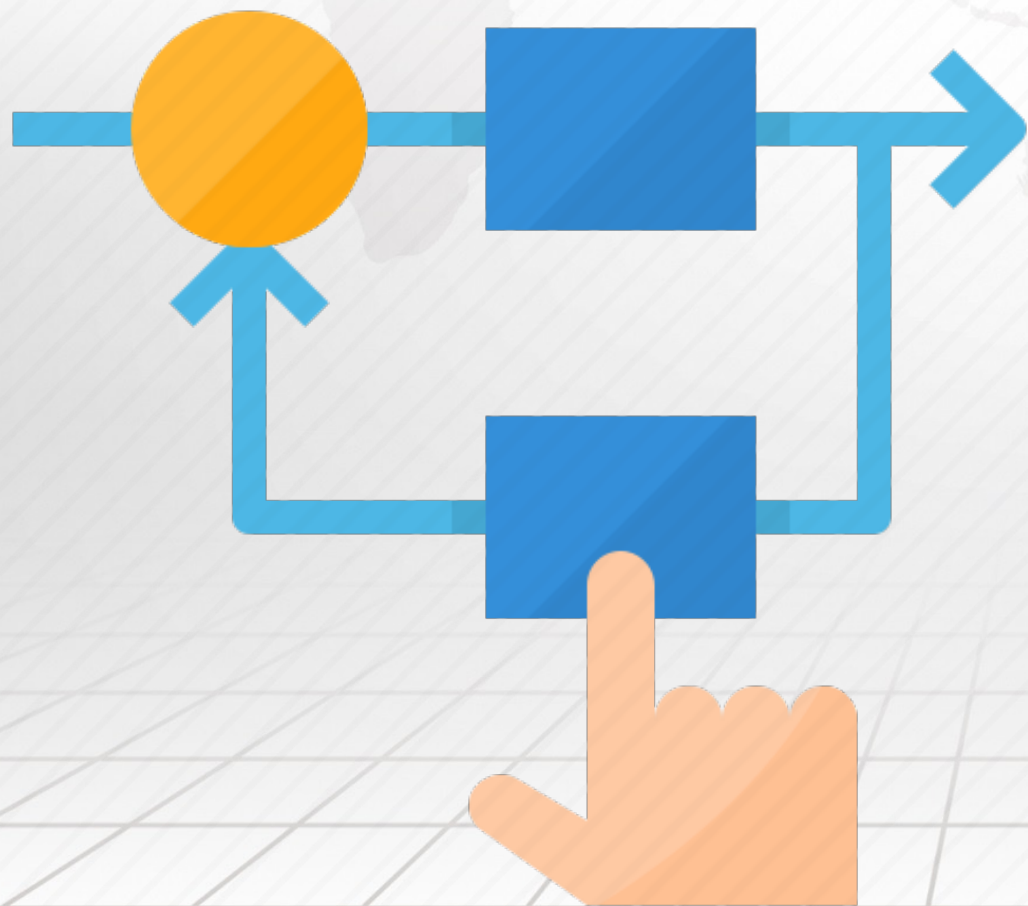
自动控制原理

第一章	第二章	第三章	第四章	第五章	第六章
基本概念	数学模型	时域分析法	根轨迹法	频域分析法	校正方法

16. 频率特性：换个角度看控制



学习目标



01

能复述频率响应的含义

重

掌握典型环节的对数幅频特性

难

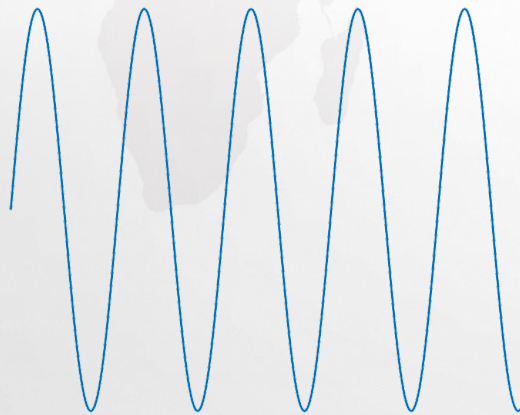
能求解频率响应的时域表达式

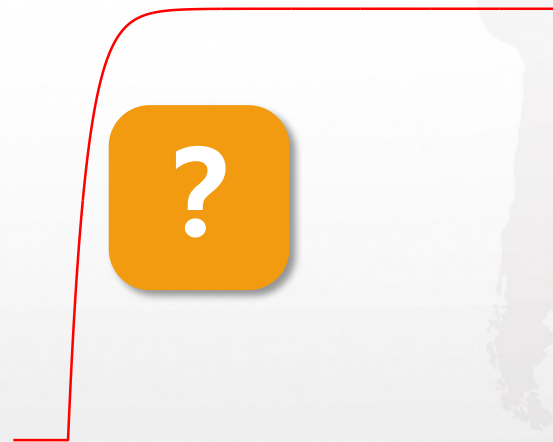
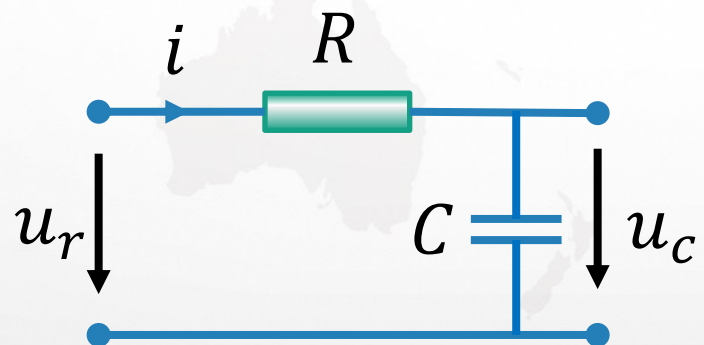
04

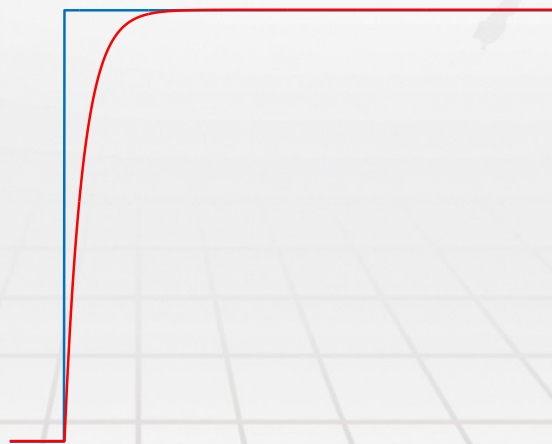
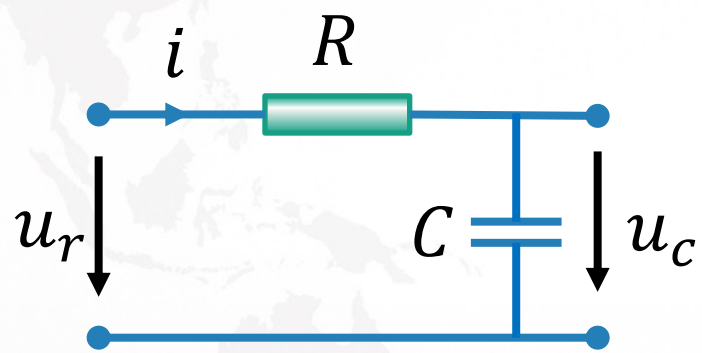
能够运用仿真验证频率响应

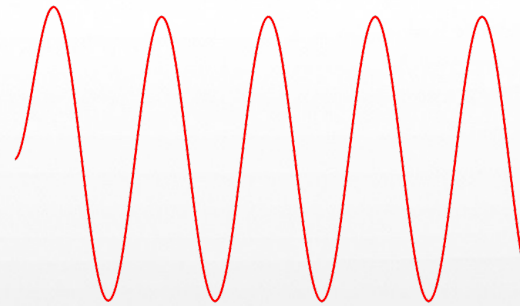
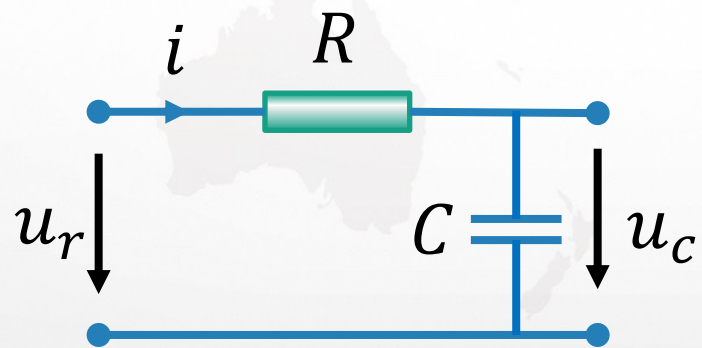
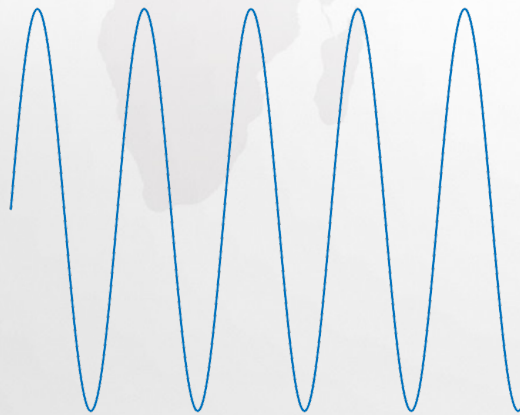


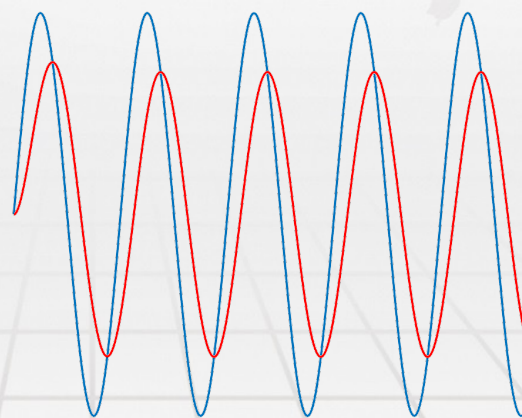
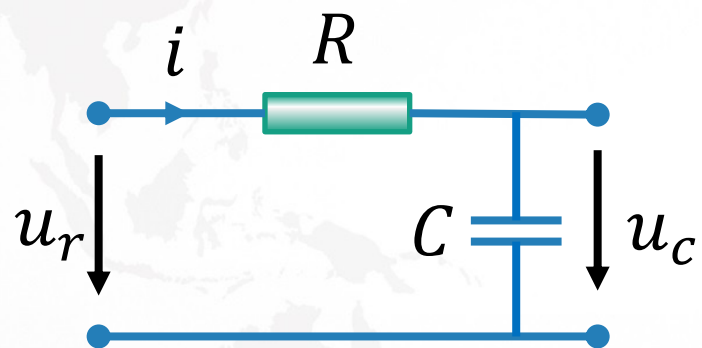








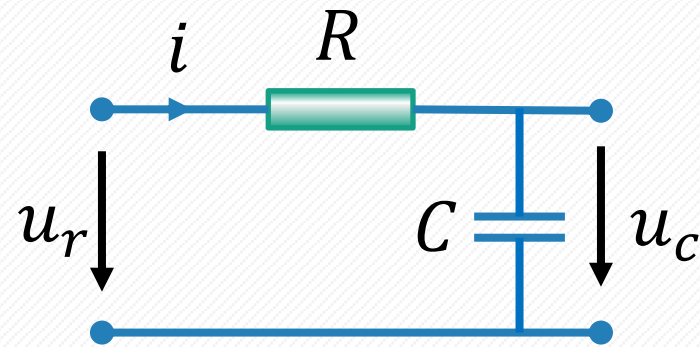




正弦信号作用下的响应

?

输入信号为 $u_r(t) = A \sin \omega t$ 时, 求 $u_c(t)$?



$$G(s) = \frac{U_c(s)}{U_r(s)} = \frac{1}{CRs+1} \xrightarrow{T=CR} \frac{1}{Ts+1} = \frac{1/T}{s+1/T}$$

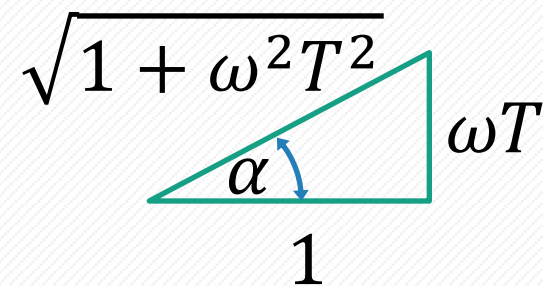
$$U_c(s) = \frac{1/T}{s+1/T} \cdot \frac{A\omega}{s^2+\omega^2} \xrightarrow{\text{部分分式分解}} \frac{C_0}{s+1/T} + \frac{C_1s+C_2}{s^2+\omega^2}$$

$$C_0 = \lim_{s \rightarrow -1/T} \frac{A\omega/T}{s^2+\omega^2} = \frac{A\omega T}{1+\omega^2 T^2} \quad C_1 = \frac{-A\omega T}{1+\omega^2 T^2} \quad C_2 = \frac{A\omega}{1+\omega^2 T^2}$$

$$U_c(s) = \frac{A\omega T}{1+\omega^2 T^2} \cdot \frac{1}{s+1/T} + \frac{A}{\sqrt{1+\omega^2 T^2}} \left[\frac{1}{\sqrt{1+\omega^2 T^2}} \cdot \frac{\omega}{s^2+\omega^2} - \frac{\omega T}{\sqrt{1+\omega^2 T^2}} \cdot \frac{s}{s^2+\omega^2} \right]$$

正弦信号作用下的响应

$$U_c(s) = \frac{A\omega T}{1+\omega^2 T^2} \cdot \frac{1}{s+1/T} + \frac{A}{\sqrt{1+\omega^2 T^2}} \left[\frac{1}{\sqrt{1+\omega^2 T^2}} \cdot \frac{\omega}{s^2+\omega^2} - \frac{\omega T}{\sqrt{1+\omega^2 T^2}} \cdot \frac{s}{s^2+\omega^2} \right]$$

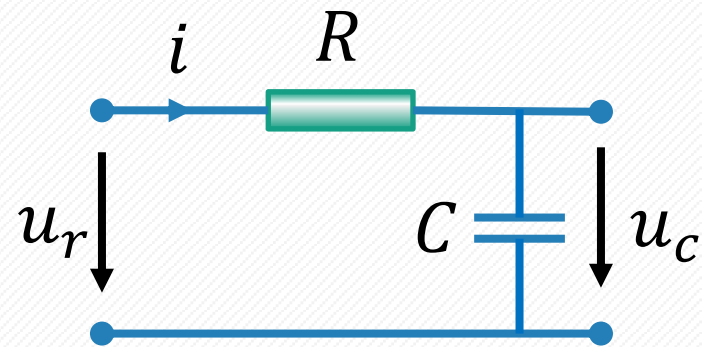


$$u_c(t) = \frac{A\omega T}{1+\omega^2 T^2} e^{-t/T} + \frac{A}{\sqrt{1+\omega^2 T^2}} [\sin \omega t \cdot \cos \alpha - \cos \omega t \cdot \sin \alpha]$$

积化和差： $\sin \omega t \cos \alpha = \frac{\sin(\omega t + \alpha) + \sin(\omega t - \alpha)}{2}$, $\cos \omega t \sin \alpha = \frac{\sin(\omega t + \alpha) - \sin(\omega t - \alpha)}{2}$

$$u_c(t) = \frac{A\omega T}{1+\omega^2 T^2} e^{-t/T} + \frac{A}{\sqrt{1+\omega^2 T^2}} \sin(\omega t - \arctan \omega T) \quad \gg \quad \text{指数衰减+正弦}$$

正弦信号作用下的响应



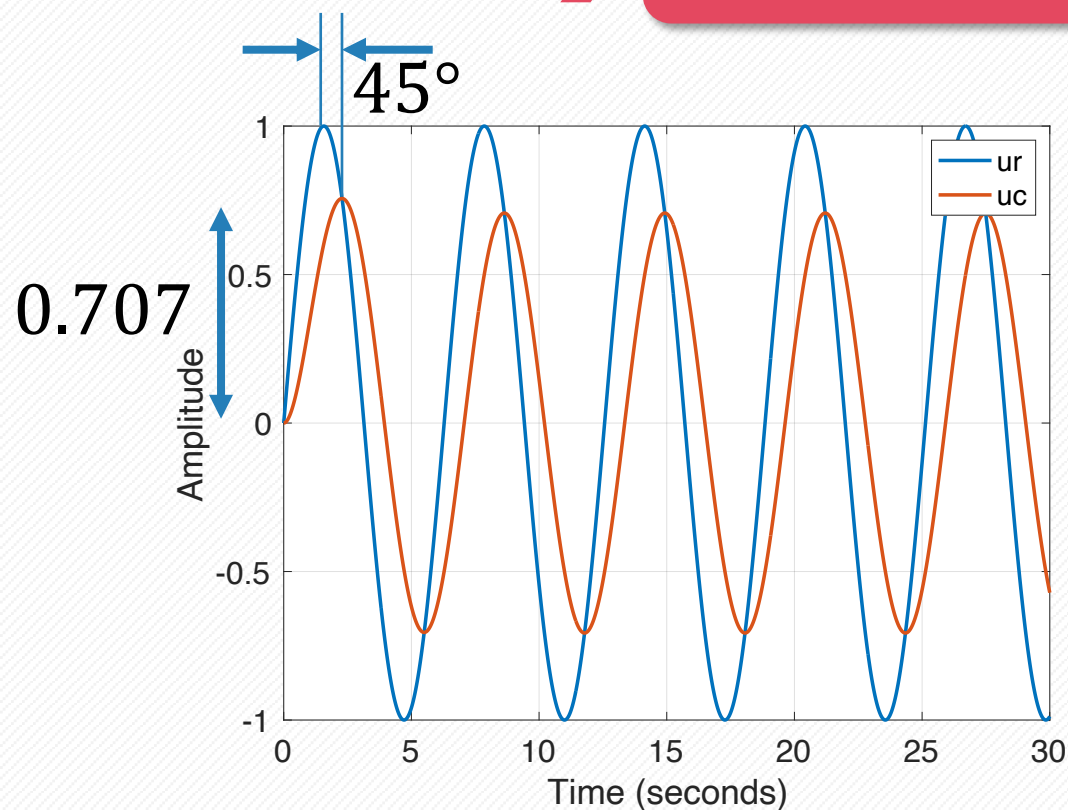
$$u_c(t) = \frac{A\omega T}{1+\omega^2 T^2} e^{-t/T} + \frac{A}{\sqrt{1+\omega^2 T^2}} \sin(\omega t - \arctan \omega T)$$

$$A = 1, \omega = 1, T = 1$$

$$= \frac{1}{2} e^{-t} + \frac{1}{\sqrt{2}} \sin(t - 45^\circ)$$



指数衰减+正弦

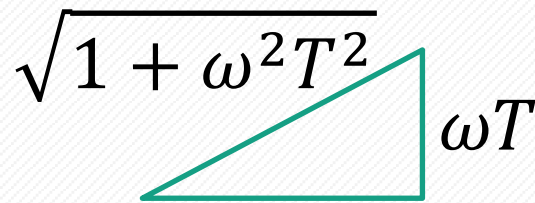


频率特性 $G(j\omega)$ 的定义

正弦响应的稳态部分

$$u_r(t) = A \sin \omega t$$

$$c_s(t) = \frac{A}{\sqrt{1+\omega^2 T^2}} \sin(\omega t - \arctan \omega T)$$

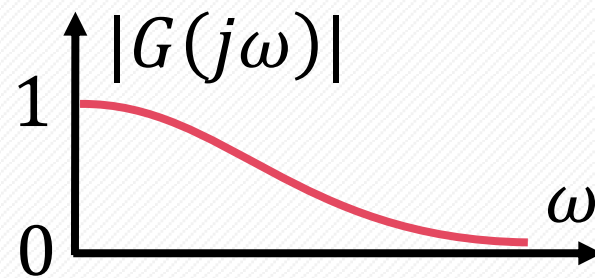


定义一

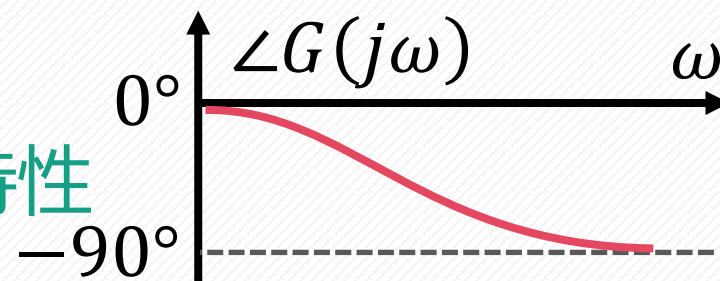
$$G(j\omega) = |G(j\omega)| \angle G(j\omega)$$

$$\begin{cases} |G(j\omega)| = \frac{|c_s(t)|}{|r(t)|} = \frac{1}{\sqrt{1+\omega^2 T^2}} \\ \angle G(j\omega) = \angle c_s(t) - \angle r(t) = -\arctan \omega T \end{cases}$$

幅频特性



相频特性



定义二

$$G(j\omega) = G(s) \Big|_{s=j\omega}$$

$$\frac{1}{\sqrt{1+\omega^2 T^2}} \angle -\arctan \omega T = \left| \frac{1}{1+j\omega T} \right| \angle \frac{1}{1+j\omega T} = \frac{1}{1+j\omega T} = \frac{1}{Ts+1} \Big|_{s=j\omega}$$

定义三

$$G(j\omega) = \frac{C(j\omega)}{R(j\omega)}$$

求稳态频率响应

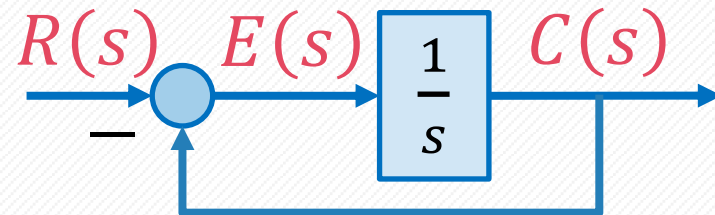
$$r(t) = 3 \sin(2t + 30^\circ)$$

$$\Phi(s) = \frac{1}{s+1} \Rightarrow \Phi(j\omega) = \frac{1}{1+j\omega}$$

$$\begin{cases} |\Phi(s)| = \left| \frac{1}{1+j\omega} \right| = \frac{1}{\sqrt{1+\omega^2}} \xrightarrow{\omega=2} \frac{1}{\sqrt{5}} = \frac{|c_s(t)|}{|r(t)|} = \frac{|c_s(t)|}{3} \\ \angle \Phi(j\omega) = -\arctan \omega \xrightarrow{\omega=2} -63.4^\circ = \angle c_s(t) - \angle r(t) = \angle c_s(t) - 30^\circ \end{cases}$$

$$\begin{cases} |c_s(t)| = \frac{3}{\sqrt{5}} \\ \angle c_s(t) = -63.4^\circ + 30^\circ = -33.4^\circ \end{cases}$$

$$\Rightarrow c_s(t) = \frac{3}{\sqrt{5}} \sin(2t - 33.4^\circ)$$



求稳态频率响应

代码

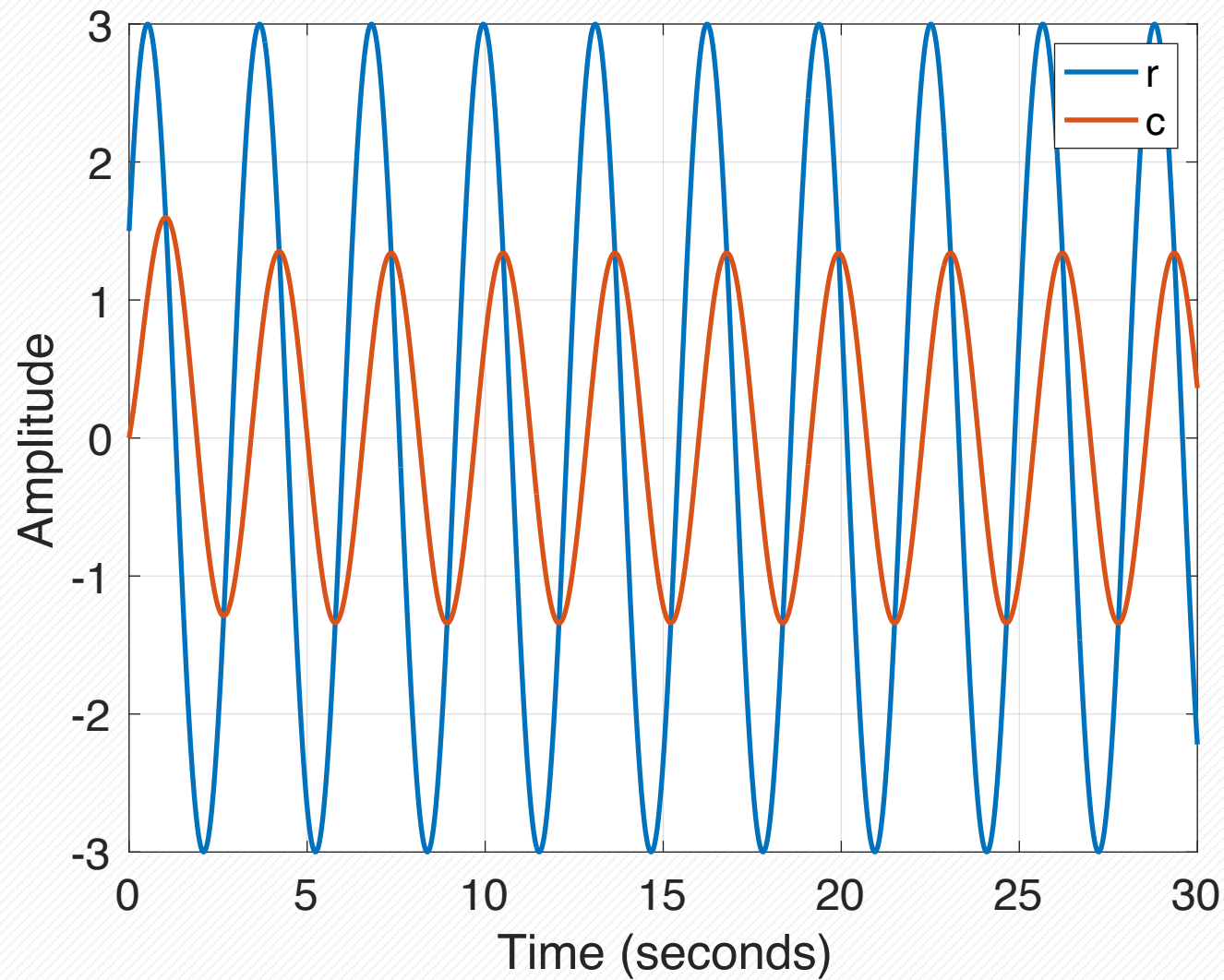
```
t = linspace(0,30,1000);  
A = 3;  
omega = 2;  
r = A*sin(omega*t+30/360*2*pi);  
G = tf(1,[1,0]);  
Phi = feedback(G,1);  
[c,t] = lsim(Phi,r,t);  
plot(t,[r',c], 'LineWidth',2);  
xlabel('Time (seconds)')  
ylabel('Amplitude')  
legend({'r','c'})  
grid
```

```
% 生成1000个时间点  
% 幅值  
% 频率  
% 输入正弦信号  
% 开环传函  
% 单位负反馈  
% 时域响应仿真  
% 绘图
```

求稳态频率响应

$$r(t) = 3 \sin(2t + 30^\circ)$$

$$\Rightarrow c_s(t) = \frac{3}{\sqrt{5}} \sin(2t - 33.4^\circ)$$



频率特性 $G(j\omega)$ 的定义

系统的频率特性 $G(j\omega) = |G(j\omega)| \angle G(j\omega)$

反映了在正弦信号作用下，系统的稳态响应与输入正弦信号的关系。

幅频特性

$$|G(j\omega)| = \frac{M(\omega)}{A}$$

反映系统在不同频率正弦信号作用下，稳态输出信号与输入信号幅值的比值，及系统的放大(或衰减)特性。

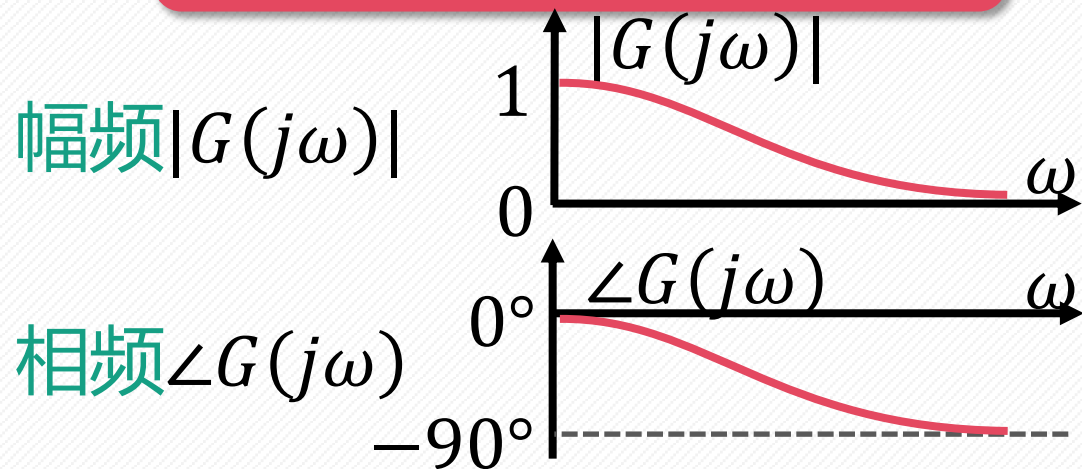
相频特性

$$\angle G(j\omega) = \phi(\omega) = \arctan \left[\frac{\text{Im}[G(j\omega)]}{\text{Re}[G(j\omega)]} \right]$$

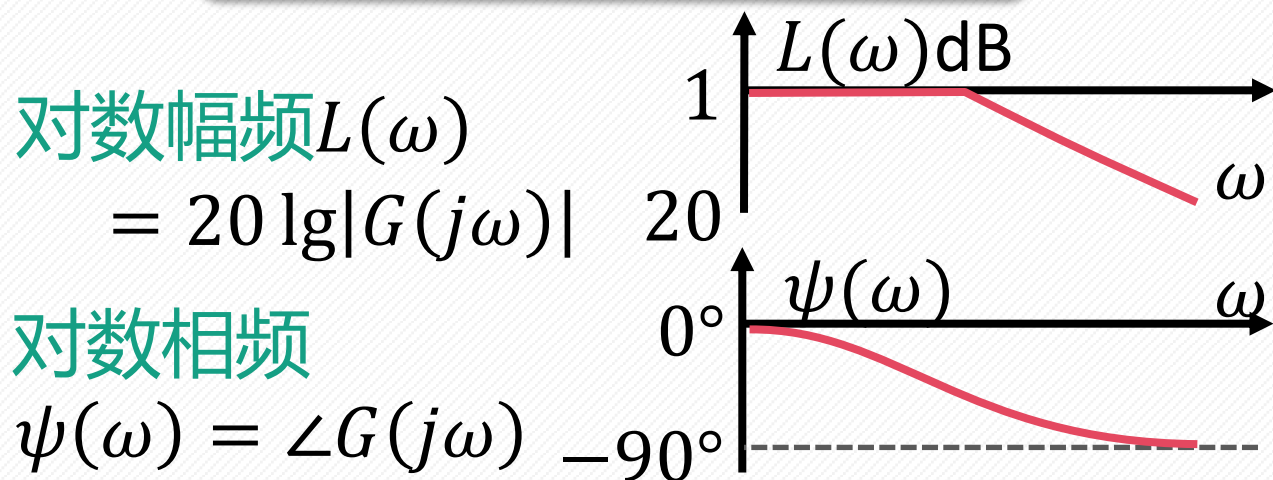
反映系统在不同频率正弦信号作用下，稳态输出信号相对于输入信号的相移。

频率特性 $G(j\omega)$ 的图解表示

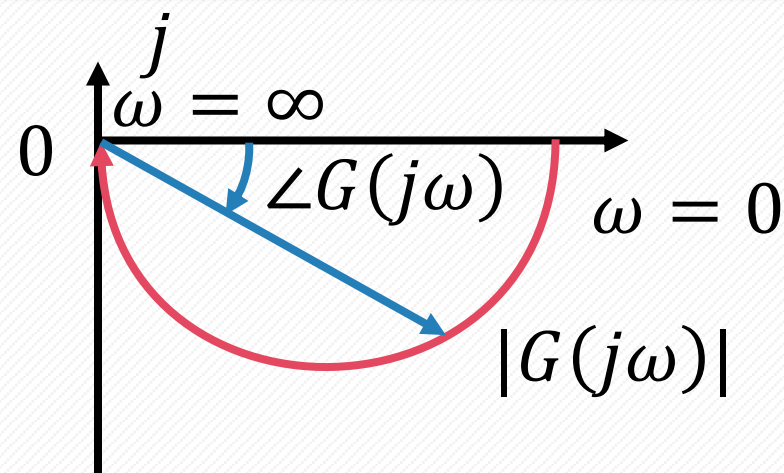
I. 频率特性



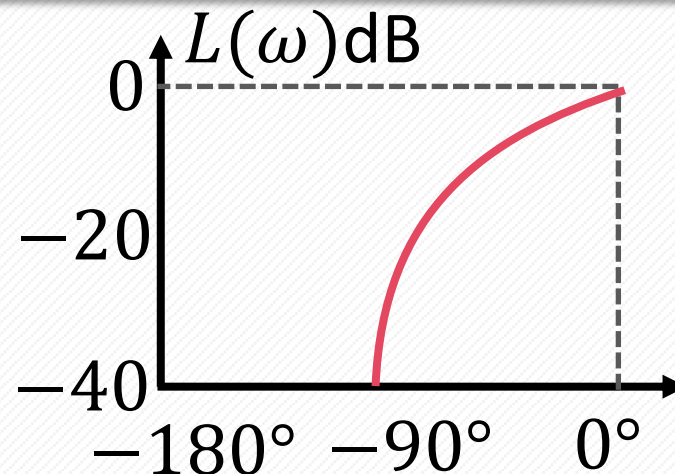
III. 对数频率特性(Bode)



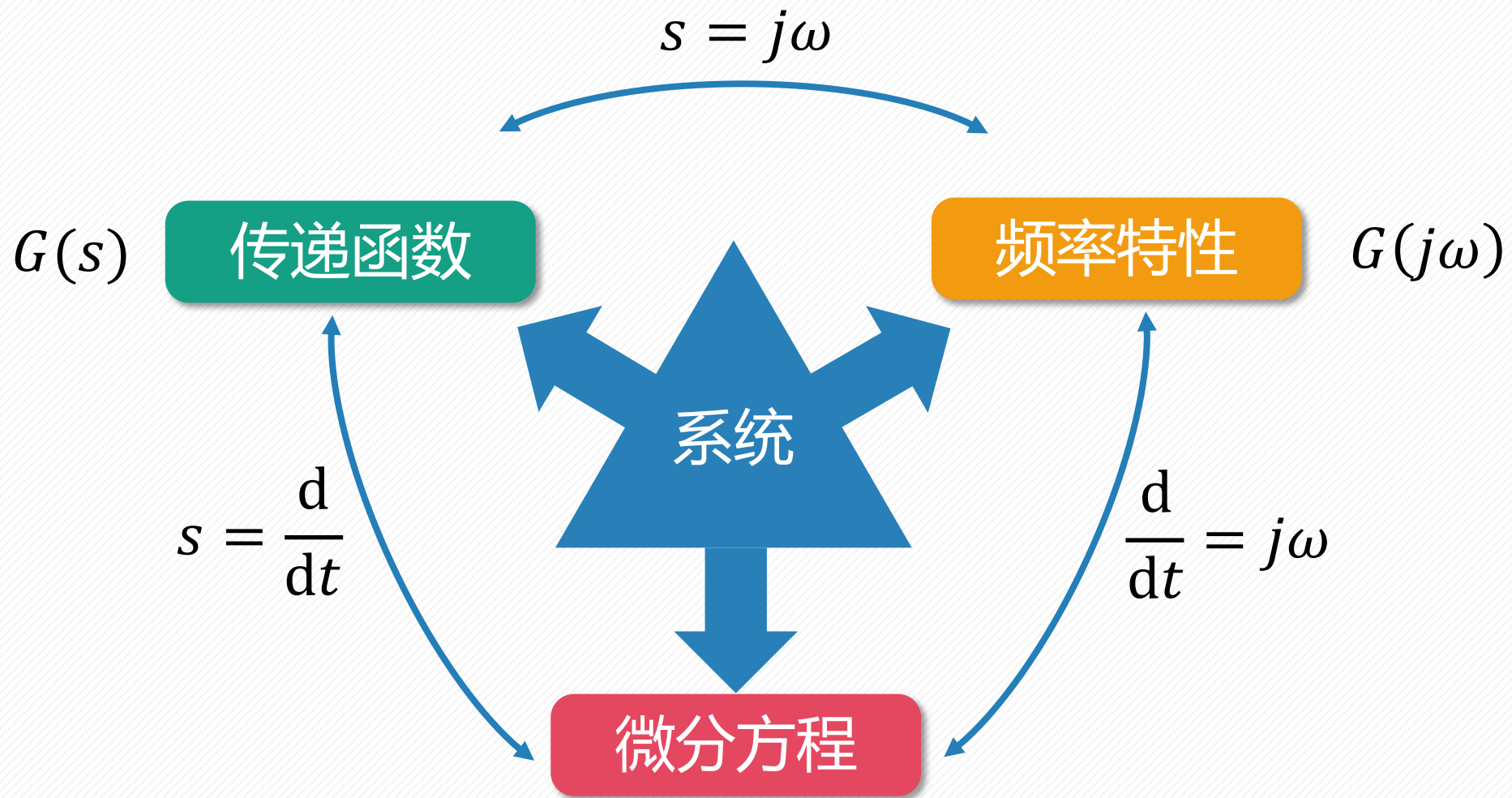
II. 幅相特性(Nyquist)、极坐标



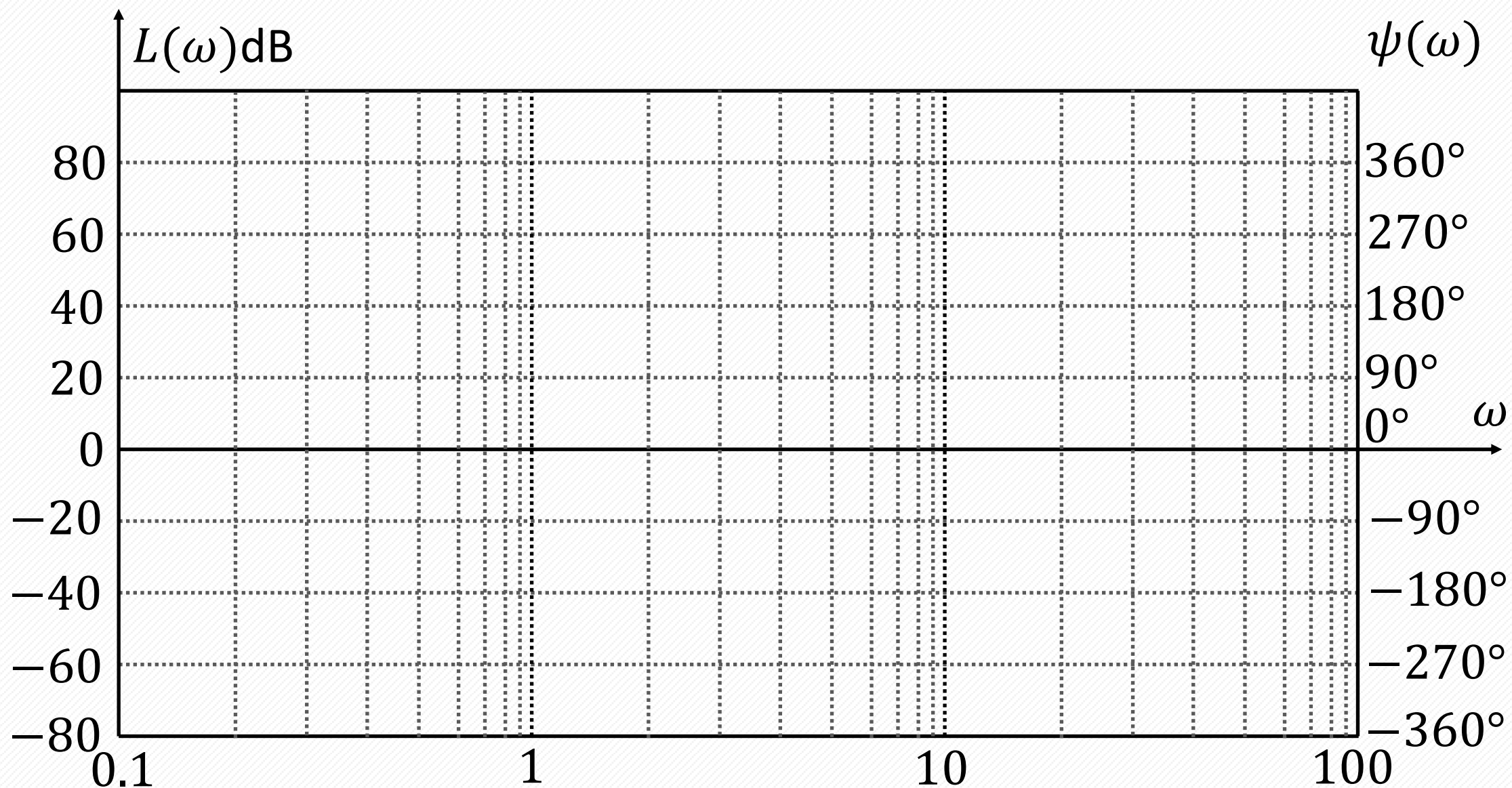
IV. 对数幅相特性(Nichols)



系统模型间的关系



对数频率特性(Bode 图)



对数频率特性(Bode 图)的特点

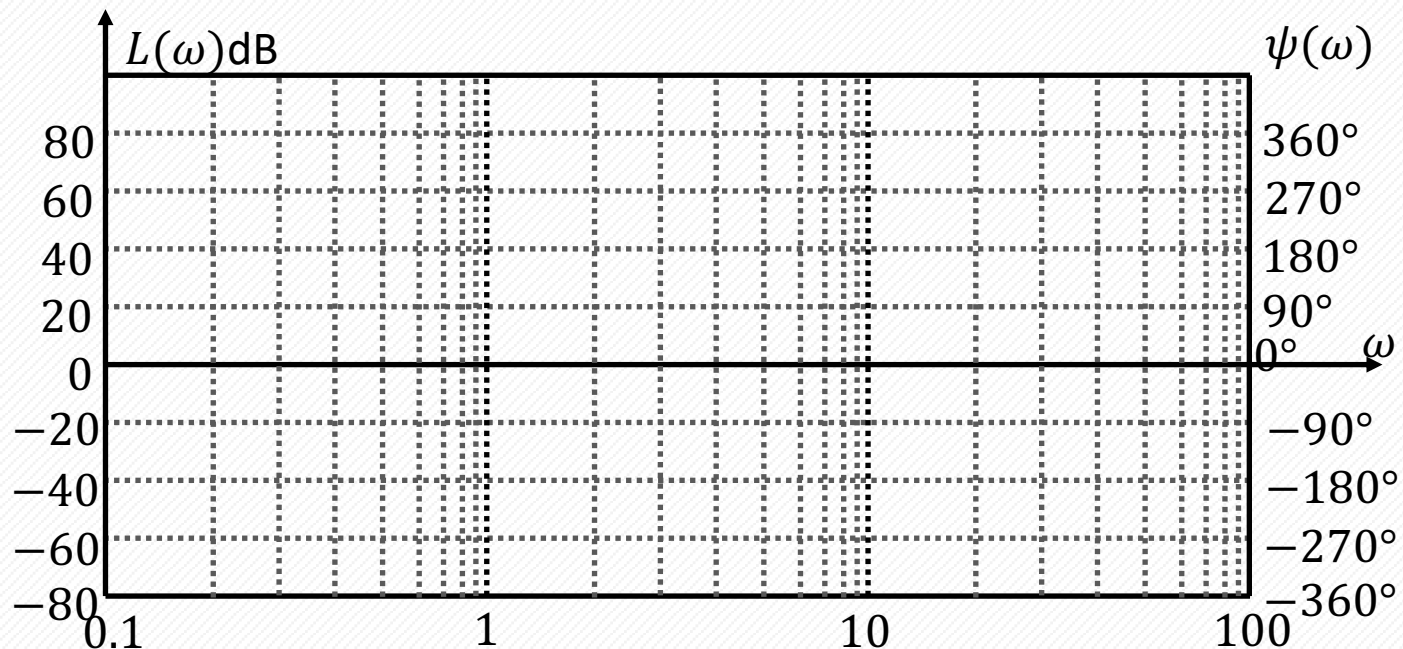
横轴

按 $\lg \omega$ 刻度, dec “十倍频程”
按 ω 标定, 等距等比

纵轴

$L(\omega) = 20 \lg |G(j\omega)|$ dB “分贝”

特点



幅值相乘 = 对数相加, 便于叠加作图;

可在大范围内表示频率特性;

使各个频段(分别关系到系统各种性能)都可以得到同等重视;

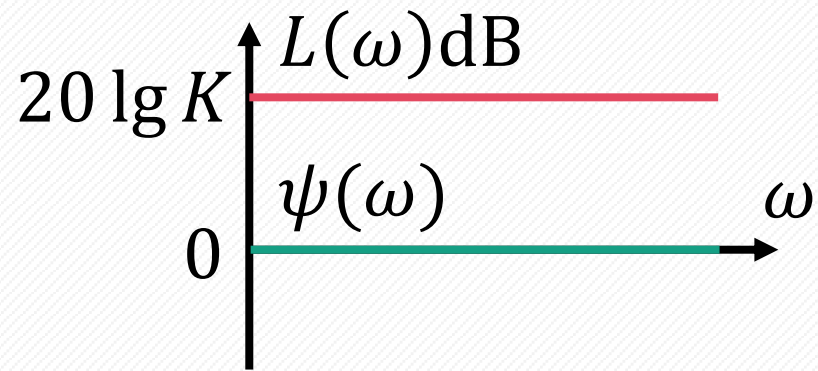
(非直线)特性曲线可以绘制渐近对数幅频特性, 进一步简化绘制过程;

最小相位系统, 可由对数幅频特性曲线得到系统的传递函数。

典型环节的 Bode 图

01 比例环节

$$G(j\omega) = K, \begin{cases} L(\omega) = 20 \lg K \\ \psi(\omega) = 0^\circ \end{cases}$$

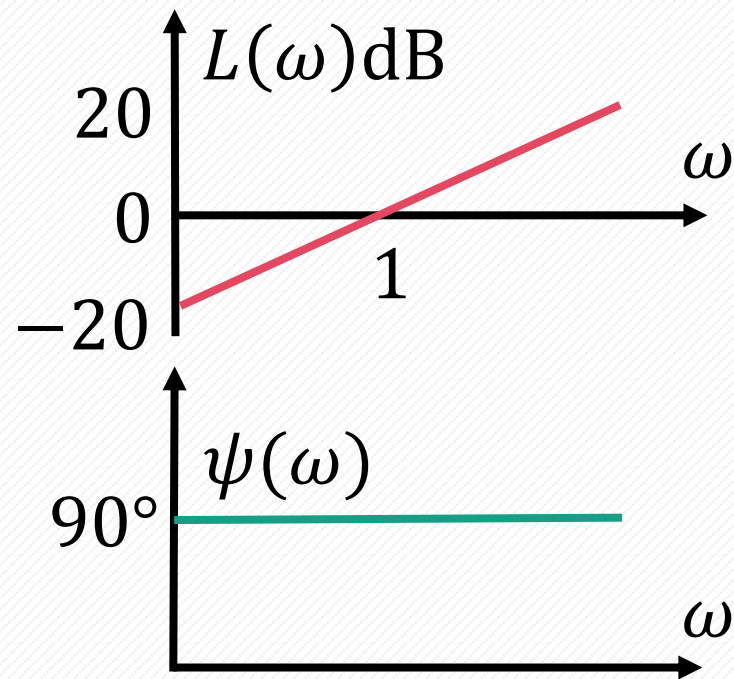


02 微分环节

$$G(j\omega) = j\omega, \begin{cases} L(\omega) = 20 \lg \omega \\ \psi(\omega) = 90^\circ \end{cases}$$

$L(\omega)$ 是一条斜率为 $+20\text{dB/dec}$ 过 $(1,0)$ 点的直线, 记作 $[+20]$

$$G(j\omega) = (j\omega)^\alpha, \begin{cases} L(\omega) = 20\alpha \lg \omega \\ \psi(\omega) = \alpha \times 90^\circ \end{cases}$$



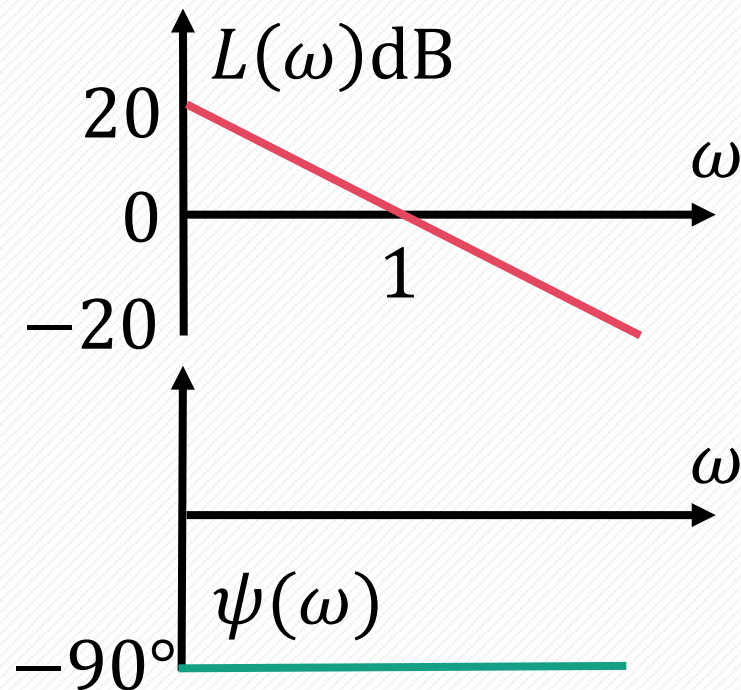
典型环节的 Bode 图

03 积分环节

$$G(j\omega) = \frac{1}{j\omega}, \begin{cases} L(\omega) = -20 \lg \omega \\ \psi(\omega) = -90^\circ \end{cases}$$

$L(\omega)$ 是一条斜率为 -20dB/dec ,
过 $(1,0)$ 点的直线, 记作 $[-20]$ 。

$$G(j\omega) = \left(\frac{1}{j\omega}\right)^\alpha, \begin{cases} L(\omega) = -20\alpha \lg \omega \\ \psi(\omega) = -\alpha \times 90^\circ \end{cases}$$



典型环节的 Bode 图

04 惯性环节

$$G(j\omega) = \frac{1}{1 + j\omega T}, \begin{cases} L(\omega) = -20 \lg \sqrt{1 + \omega^2 T^2} \\ \psi(\omega) = -\arctan \omega T \end{cases}$$

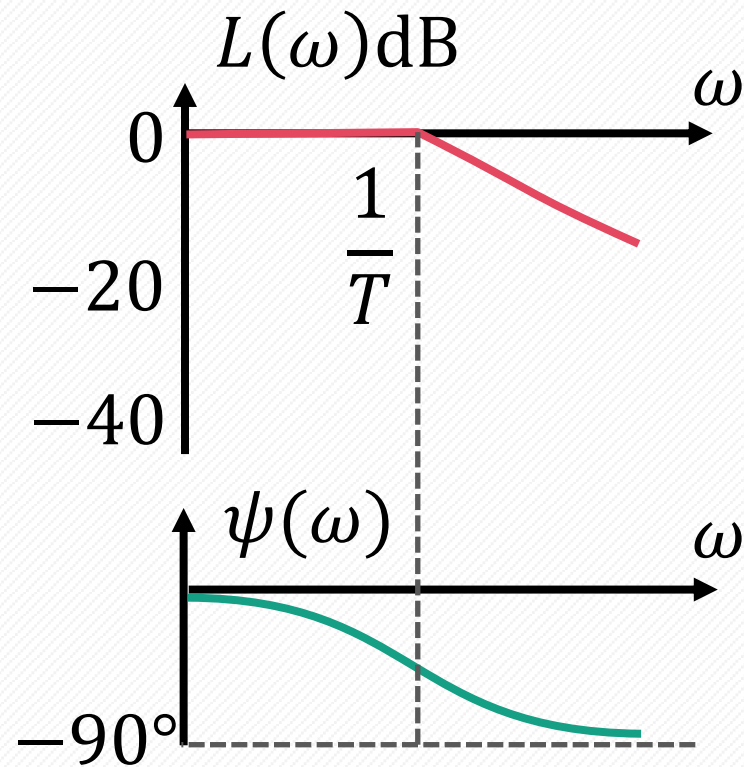
渐近线

惯性环节可由一组折线组成的渐近线近似

交接频率

渐近线各近似线段相交的交点频率称为交接频率。交接频率处误差最大可达

$$\left. -20 \lg \sqrt{1 + \omega^2 T^2} \right|_{\omega = \frac{1}{T}} = -20 \lg \sqrt{2} = -3(\text{dB})$$



- 误差曲线对称于交接频率
- 误差主要位于交接频率上下十倍频程范围内

典型环节的 Bode 图

05 一阶微分

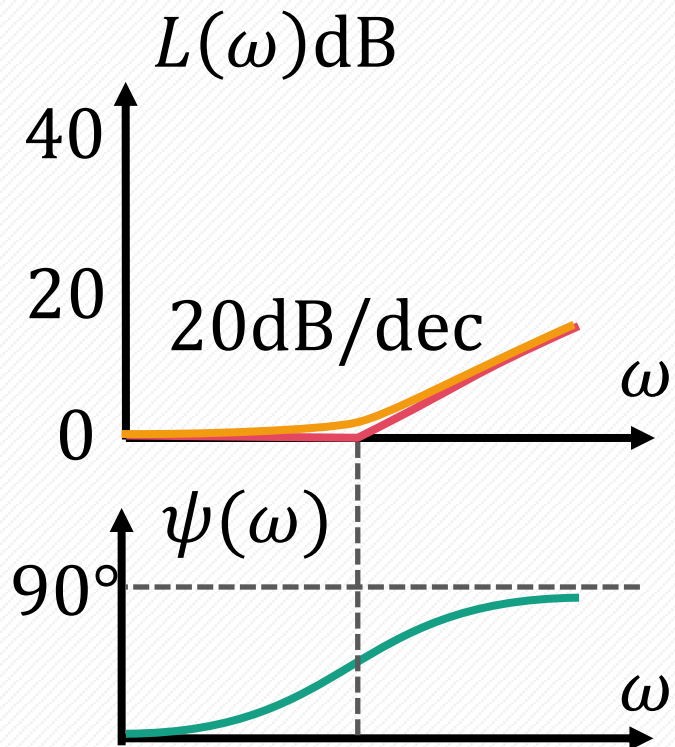
$$G(j\omega) = 1 + j\omega\tau, \begin{cases} L(\omega) = 20 \lg \sqrt{1 + \omega^2\tau^2} \\ \psi(\omega) = \arctan \omega\tau \end{cases}$$

交接频率

$$1/\tau$$

交接频率处误差最大可达

$$20 \lg \sqrt{1 + \omega^2\tau^2} \Big|_{\omega=\frac{1}{\tau}} = 20 \lg \sqrt{2} = 3(\text{dB})$$



- 误差曲线对称于交接频
- 误差主要位于交接频率上下十倍频程范围内

典型环节的 Bode 图

06 震荡环节

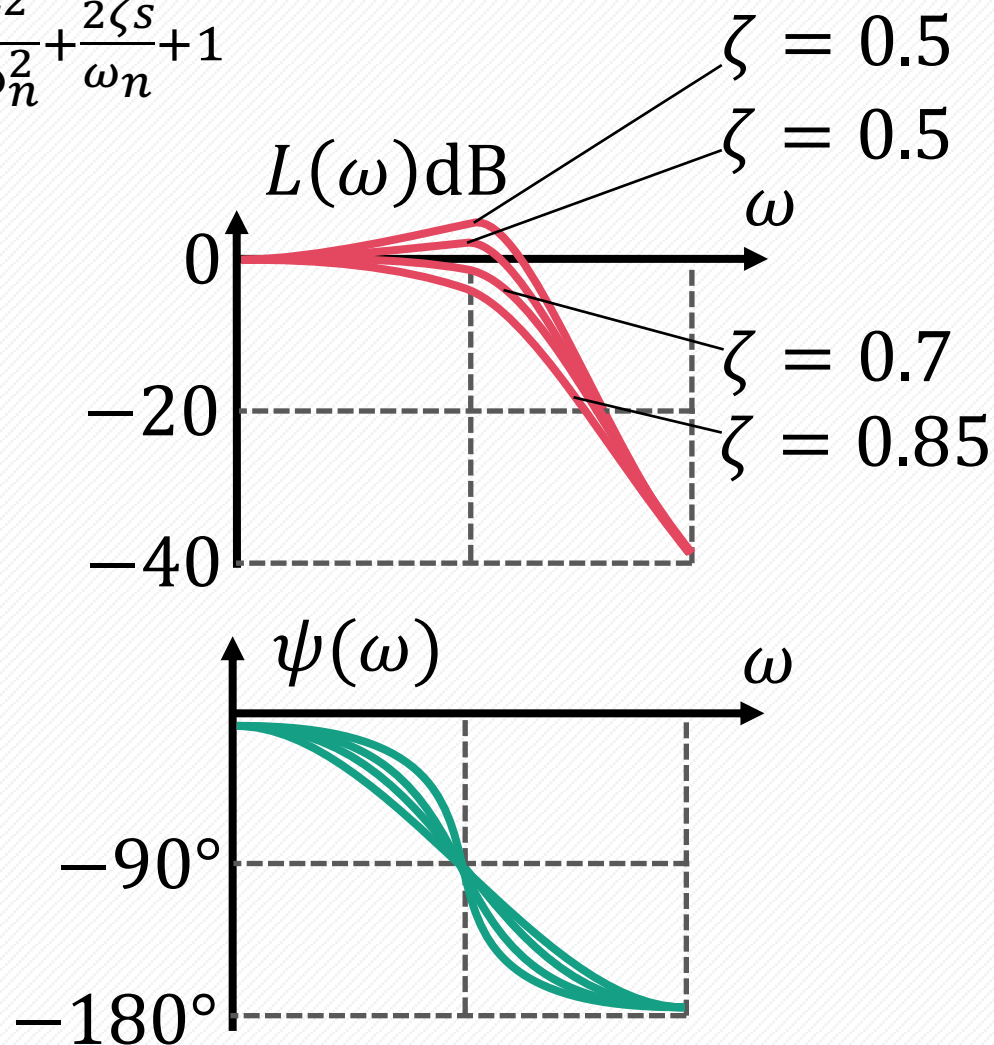
$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{1}{\frac{s^2}{\omega_n^2} + \frac{2\zeta s}{\omega_n} + 1}$$

$$G(j\omega) = \frac{1}{1 - \frac{\omega^2}{\omega_n^2} + j2\zeta\frac{\omega}{\omega_n}}$$

$$\begin{cases} L(\omega) = -20 \lg \sqrt{\left[1 - \frac{\omega^2}{\omega_n^2}\right]^2 + \left[2\zeta\frac{\omega}{\omega_n}\right]^2} \\ \psi(\omega) = \arctan \left[\left(2\zeta\frac{\omega}{\omega_n}\right) / \left(1 - \frac{\omega^2}{\omega_n^2}\right) \right] \end{cases}$$

$$\frac{\omega}{\omega_n} \ll 1 \begin{cases} L(\omega) \approx 0 \\ \psi(\omega) \approx 0 \end{cases}$$

$$\frac{\omega}{\omega_n} \gg 1 \begin{cases} L(\omega) \approx -40 \lg(\omega/\omega_n) \\ \psi(\omega) \approx 180^\circ \end{cases}$$



总结

01 频率响应

正弦响应的稳态部分

02 BODE图

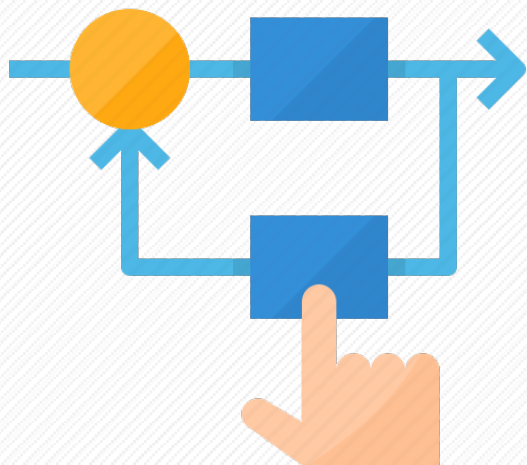
频率取对数，幅值取对数后乘20得到频率响应图

03 典型环节

比例，积分，微分，惯性，一阶微分，振荡

04 10倍频程

频率变化十倍在幅频特性上对应的范围



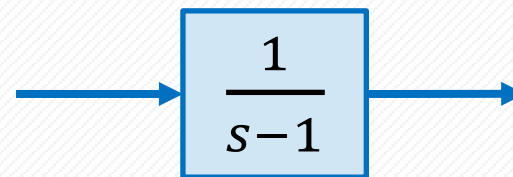
横看成岭侧成峰

多视角分析

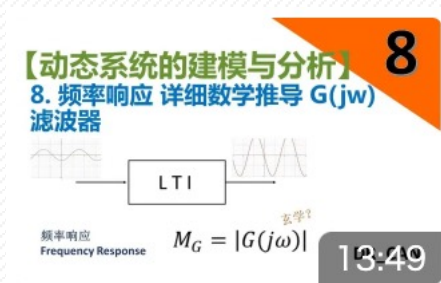
课后思考

?

试分析不稳定系统是否也具有稳定的频率响应，如果没有，分析其具有的形态，并通过仿真验证你的分析。



课外拓展及预习要求



DR_CAN频率响应_详细数学推导 $G(jw)$:
https://www.bilibili.com/video/BV1Qb411W7i2?s_hare_source=copy_web

- [1] Rake, H. (1979). Step response and frequency response methods. IFAC Proceedings Volumes, 12(8), 519-526.
- [2] Wang, W., Yao, W., Chen, C., Deng, X., & Liu, Y. (2020). Fast and accurate frequency response estimation for large power system disturbances using second derivative of frequency data. IEEE Transactions on Power Systems, 35(3), 2483-2486.

16. 频率特性：换个角度看控制

感谢聆听！

